Statistics 652
Problem Set 3 - Fall, 2026 *Note: This problem set may be updated.*

3.1. Given a scatter plot of x_1 versus x_2, discuss how to plot the values of $y = \mathbf{a}^T\mathbf{x} = a_1x_1 + a_2x_2$.

3.2. Let $\underline{\overline{\mathbf{X}}} = [\mathbf{X}_1 \cdots \mathbf{X}_n]^{\mathrm{T}}$ represents the $n \times p$ data matrix for the sample $\mathbf{X}_1, \ldots, \mathbf{X}_n \in \mathrm{R}^p$. Show that the sample mean vector and sample covariance matrix are respectively $\overline{\mathbf{X}} = n^{-1}\underline{\overline{\mathbf{X}}}^{\mathrm{T}}\mathbf{1}_n$ and $\mathbf{S}_n = n^{-1}\underline{\overline{\mathbf{X}}}^{\mathrm{T}}\mathbf{H}_n\underline{\overline{\mathbf{X}}}$, where $\mathbf{1}_n$ is the n-dimensional vector of all ones, and $\mathbf{H}_n = \mathbf{I}_n - n^{-1}\mathbf{1}_n\mathbf{1}_n^{\mathrm{T}}$, which corresponds to the orthogonal projection onto the space orthogonal to $\mathbf{1}_n$ in R^n.

3.3. For the random vector $\mathbf{X} \in \mathrm{R}^p$, let $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A} \in \mathrm{R}^{q\times p}$, i.e. a $q \times p$ matrix, $\mathbf{b} \in \mathrm{R}^q$. Show (i) $\boldsymbol{\mu}_Y = \mathbf{A}\boldsymbol{\mu}_X + \mathbf{b}$, and (ii) $\boldsymbol{\Sigma}_Y = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^{\mathrm{T}}$.

3.4. For $\mathbf{X} \in \mathrm{R}^p$, let $Y_1 = \mathbf{a}^T\mathbf{X}$ and $Y_2 = \mathbf{b}^T\mathbf{X}$, with $\mathbf{a}, \mathbf{b} \in \mathrm{R}^p$. Show that the covariance between Y_1 and Y_2 is given by $\sigma_{Y_1,Y_2} = \mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}_X\mathbf{b} = \mathbf{b}^{\mathrm{T}}\boldsymbol{\Sigma}_X\mathbf{a}$.

3.5. Consider an equation in $\mathbf{x} \in \mathrm{R}^p$ of the form $(\mathbf{x} - \boldsymbol{\mu})^{\mathrm{T}}\mathbf{A}(\mathbf{x} - \boldsymbol{\mu}) = r^2$.

i) State necessary and sufficient conditions on $\mathbf{A}$ for this equation to be an ellipse.

ii) Under these conditions, find the center, the axes and the corresponding lengths of the axes of the ellipse.

ii) Find the equation of the two dimensional ellipse centered at $(1, 2)$, with axes of lengths 4 and 2, and whose major axis is 30° from horizontal.

3.6. Show that a nonsingular linear transformation maps a circle into an ellipse. That is, if $\mathbf{B}$ is a $p \times p$ nonsingular matrix, $\mathbf{b} \in \mathrm{R}^p$ and $\mathbf{y} = \mathbf{B}\mathbf{x} + \mathbf{b}$, then

$$\{\mathbf{x} \in \mathrm{R}^p \mid (\mathbf{x} - \mathbf{c})^T(\mathbf{x} - \mathbf{c}) = r^2\} = \{\mathbf{y} \in \mathrm{R}^p \mid (\mathbf{y} - \boldsymbol{\mu})^T\mathbf{A}(\mathbf{y} - \boldsymbol{\mu}) = r^2\},$$

for some $\mathbf{A} > \mathbf{0}$ and $\boldsymbol{\mu}$. Also, find the values of $\mathbf{A}$ and $\boldsymbol{\mu}$.

3.7. Show that if the $p \times p$ matrix $\mathbf{A} \geq \mathbf{0}$ is not symmetric, then $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T) \geq \mathbf{0}$ and $\mathbf{A}_s$ is symmetric. Furthermore, for any $\mathbf{a} \in \mathrm{R}^p$, $\mathbf{a}^T\mathbf{A}_s\mathbf{a} = \mathbf{a}^T\mathbf{A}\mathbf{a}$.

3.8. Show that if $\mathbf{A} > \mathbf{0}$ then $\mathbf{A}^{-1}$ exists and $\mathbf{A}^{-1} > \mathbf{0}$.

3.9. i) Show that a covariance matrix $\boldsymbol{\Sigma}$ is positive semi-definite.

ii) Show that if $\boldsymbol{\Sigma} > 0$ then the correlation matrix $\mathbf{R} > \mathbf{0}$.

iii) Show that if the covariance matrix of $\mathbf{X} \in \mathrm{R}^q$ is singular, then there exists a least one linear combination $\mathbf{a}^{\mathrm{T}}\mathbf{X}$ with zero variance.

iv) Show that if the covariance matrix of $\mathbf{X} \in \mathrm{R}^q$ is singular and $\mathbf{X}$ is has a continuous distribution, then the distribution of $\mathbf{X}$ is singular.

v) Give an example of a singular distribution which does not have a singular covariance matrix.

3.10 For $\boldsymbol{\Sigma} > \mathbf{0}$, show

$$\sup_{\mathbf{a}\in\mathrm{R}^p, \mathbf{a}\neq\mathbf{0}} \frac{|\mathbf{a}^{\mathrm{T}}\mathbf{x} - \mathbf{a}^{\mathrm{T}}\boldsymbol{\mu}|}{\sqrt{\mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{a}}} = D(\mathbf{x}), \text{ where } D^2(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$$

with the supremum being obtained only whenever $\mathbf{a} \propto \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$.

3.11 For $\mathbf{\Sigma} > \mathbf{0}$, show

$$\bigcap_{\mathbf{a}\in \mathrm{R}^p, \mathbf{a}\neq \mathbf{0}} \{\mathbf{x} \in \mathrm{R}^p \mid \mathbf{a}^{\mathrm{T}}\mathbf{x} \in \mathbf{a}^{\mathrm{T}}\boldsymbol{\mu} \pm k\sqrt{\mathbf{a}^{\mathrm{T}}\mathbf{\Sigma}\mathbf{a}}\} = \{\mathbf{x} \in \mathrm{R}^p \mid (\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}) \leq k^2\}.$$

3.12. Show that the orthogonal projection of an ellipse in R^p onto any $q < p$ dimensional subspace is an ellipse in R^q. More specifically, show

$$\bigcap_{\mathbf{A}\in\mathcal{A}_q} \{\mathbf{x} \in \mathrm{R}^p \mid (\mathbf{y}-\bar{\mathbf{Y}})^T\mathbf{S}_Y^{-1}(\mathbf{y}-\bar{\mathbf{Y}}) \leq k^2, \text{ with } \mathbf{y} = \mathbf{A}\mathbf{x}\} = \{\mathbf{x} \in \mathrm{R}^p \mid (\mathbf{x}-\bar{\mathbf{X}})^T\mathbf{S}^{-1}(\mathbf{x}-\bar{\mathbf{X}}) \leq k^2\},$$

where $\mathcal{A}_q = \{\mathbf{A} \in \mathrm{R}^{q\times p} \mid \mathrm{rank}(\mathbf{A}) = q\}$.

3.13. Show that $\lambda \in \mathcal{C}$ is an algebraic eigenvalue of $\mathbf{A}_{p\times p}$, i.e. $\det(\mathbf{A} - \lambda\mathbf{I}_p) = 0$, if and only if it is a geometric eigenvalue of $\mathbf{A}$, i.e. there exists an $\mathbf{a} \in \mathcal{C}^p$ such that $\mathbf{A}\mathbf{a} = \lambda\mathbf{a}$.

3.14. For a matrix of order p, the algebraic multiplicity of an eigenvalue λ, say $m_a(\lambda)$, is its multiplicity as a solution to the pth order polynomial $\det(\mathbf{A} - \lambda\mathbf{I}_p) = 0$. The geometric multiplicity of λ, say $m_g(\lambda)$, is the dimension of its *eigenspace* $\mathcal{E}(\lambda) = \{\mathbf{a} \in \mathcal{C}^p \mid \mathbf{A}\mathbf{a} = \lambda\mathbf{a}\}$.

i) Show that $\mathcal{E}(\lambda)$ is a vector space over $\mathcal{C}$.

ii) Show $m_g(\lambda) \leq m_a(\lambda)$, and give an example for which $m_g(\lambda) < m_a(\lambda)$.

3.15. Let $\mathbf{A}$ be a real symmetric matrix of order p.

i) Show all of eigenvalues of $\mathbf{A}$ are real.

ii) Show the geometric and algebraic multiplicities of the eigenvalues of $\mathbf{A}$ are equal.

iii) Let $\lambda \in \mathrm{R}$ be an eigenvalue of $\mathbf{A}$ with multiplicity $m(\lambda)$. Show that its eigenspace $\mathcal{E}(\lambda)$ can be spanned by $m(\lambda)$ real eigenvectors of $\mathbf{A}$. This generates the *real eigenspace* $\mathcal{E}_{\mathrm{R}}(\lambda) = \{\mathbf{a} \in \mathrm{R}^p \mid \mathbf{A}\mathbf{a} = \lambda\mathbf{a}\}$. Show that $\mathcal{E}_{\mathrm{R}}(\lambda)$ is a vector space over R.

iv) Show that if $\lambda \neq \mu$ then $\mathcal{E}_{\mathrm{R}}(\lambda) \perp \mathcal{E}_{\mathrm{R}}(\mu)$ with respect to the Eucledian inner product.

vi) Denote the eigenvalues of $\mathbf{A}$ by $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. Show there exists a set of real orthonormal eigenvectors $\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_p$ such that $\mathbf{A}\boldsymbol{\gamma}_j = \lambda_j\boldsymbol{\gamma}_j$ for $j = 1, \ldots p$.

vii) Prove the spectral value decomposition for a real symmetric matrix.

viii) Let $\lambda_{(1)} > \lambda_{(2)} > \ldots > \lambda_{(k)}$ denote the eigenvalues of $\mathbf{A}$ with corresponding multiplicities $m_1, \ldots, m_k$. Thus, $m_1 + \cdots + m_k = p$. For $j = 1, \ldots k$, let $\mathcal{P}_j$ denote the corresponding *eigenprojection*, i.e. the orthogonal projection onto the real eigenspace $\mathcal{E}(\lambda_{(j)})$. Prove the following *well defined* form of the spectral value decomposition: $\mathbf{A} = \sum_{j=1}^k \lambda_{(j)}\mathcal{P}_j$.

3.16. (Rayleigh quotients.) Prove the min-max property of eigenvalues/eigenvectors of a symmetric matrix.

3.17. Show that the trace of a $p \times p$ symmetric matrix is the sum of its p eigenvalues.

3.18. Show that the eigenvalues of a symmetric matrix $\mathbf{A} \geq 0$ are non-negative, and the eigenvalues of a symmetric matrix $\mathbf{A} > 0$ are positive.

3.19. Show the equivalence of the following two definitions of a symmetric positive definite (semi-definite) matrix $\mathbf{A}$ of order p.

i) $\mathbf{A} > 0$ (≥ 0) if and only if $\mathbf{a}^T\mathbf{A}\mathbf{a} > 0$ (≥ 0) for all $\mathbf{a} \neq 0$.

ii) $\mathbf{A} \geq 0$ if and only if $\mathbf{A} = \mathbf{B}\mathbf{B}^T$ for some matrix $\mathbf{B}$ of order p, with $\mathbf{A} > 0$ if and only if $\mathbf{B}$ is nonsingular.

3.20. Suppose $\mathbf{B}\mathbf{B}^T = \mathbf{C}\mathbf{C}^T$, where $\mathbf{B}$ is $p \times r$ and $\mathbf{C}$ is $p \times m$ with $r \leq m$. Verify $\mathbf{C} = \mathbf{B}\mathbf{Q}^T$ for some $\mathbf{Q}$ such that $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_r$.

3.21. Given the spectral value decomposition for the symmetric matrix $\mathbf{A} = \mathbf{Q}\boldsymbol{\Delta}\mathbf{Q}^T$ of order q, verify $\mathbf{A}^k = \mathbf{Q}\boldsymbol{\Delta}^k\mathbf{Q}^T$.

3.22. Given a function $h : \mathrm{R} \to \mathrm{R}$, its extension to symmetric matrices or order p is defined to be $h(\mathbf{A}) = \mathbf{P}h(\Delta)\mathbf{P}^T$, where $\mathbf{A} = \mathbf{P}\boldsymbol{\Delta}\mathbf{P}^T$ is a spectral value decomposition of $\mathbf{A}$ and $h(\boldsymbol{\Delta}) = diag\{h(\lambda_1), \ldots, h(\lambda_p)\}$. Show that the function $h(\mathbf{A})$ is well defined.

3.23. Find a closed form for the partial *Neumann* sum $\sum_{k=0}^{K} \mathbf{A}^k$, where $\mathbf{A}$ is any square matrix.

3.24. Let $\mathbf{A}_1$ and $\mathbf{A}_2$ be symmetric matrices of order m. We denote $\mathbf{A}_1 > \mathbf{A}_2$ and $\mathbf{A}_1 \geq \mathbf{A}_2$ if and only if $\mathbf{A}_1 - \mathbf{A}_2 > 0$ and $\mathbf{A}_1 - \mathbf{A}_2 \geq 0$ respectively. Without using the spectral value decomposition, verify the following.

i) $\mathbf{A}_1 \geq \mathbf{A}_2 \Rightarrow trace(\mathbf{A}_1) \geq trace(\mathbf{A}_2)$.

ii) $\mathbf{A}_1 \geq \mathbf{A}_2$ and $\mathbf{A}_1 \neq \mathbf{A}_2 \Rightarrow trace(\mathbf{A}_1) > trace(\mathbf{A}_2)$.

iii) $\mathbf{A}_1 \geq \mathbf{A}_2 \Rightarrow \mathbf{C}\mathbf{A}_1C^{\mathrm{T}} \geq \mathbf{C}\mathbf{A}_2\mathbf{C}^{\mathrm{T}}$ for any $r \times m$ matrix $\mathbf{C}$.

iv) $\mathbf{A}_1 > \mathbf{A}_2 \Rightarrow \mathbf{C}\mathbf{A}_1\mathbf{C}^{\mathrm{T}} > \mathbf{C}\mathbf{A}_2\mathbf{C}^{\mathrm{T}}$ for any $r \times m$ full rank matrix $\mathbf{C}$.

3.25. Derive necessary and sufficient conditions on ρ and λ so that

$$\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} < \lambda \mathbf{I}_2.$$

3.26. Show that $(\mathbf{x}, \mathbf{y})_{\mathbf{A}} := \mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{y}$ defines an inner product on R^p when $\mathbf{A} > 0$. Also, show that any inner product on R^p must be of this form.

3.27. (a) For $\mathbf{x} \in \mathrm{R}^q$, if $g : \mathrm{R}^q \to \mathrm{R}$ such that $g(\mathbf{x}) = g(\mathbf{H}\mathbf{x})$ for all $\mathbf{H} \in \mathcal{O}(q)$, then there exists a function $h : \mathrm{R}^+ \cup \{\mathbf{0}\} \to \mathrm{R}$ such that $g(\mathbf{x}) = h(r)$, where $r = \|\mathbf{x}\|$.

(b) Show that if the function $g(\mathbf{A}) = g(\mathbf{A}\mathbf{P})$ for any $\mathbf{P} \in O(q)$, where $\mathbf{A}$ is $q \times q$, then $g(\mathbf{A}) = h(\mathbf{A}\mathbf{A}^T)$ for some function h.(b) Show that if the function $g(\mathbf{A}) = g(\mathbf{A}\mathbf{P})$ for any $\mathbf{P} \in O(q)$, where $\mathbf{A}$ is $q \times q$, then $g(\mathbf{A}) = h(\mathbf{A}\mathbf{A}^T)$ for some function h.

3.28. Let $\mathbf{Z}$ be the standardized version of $\mathbf{X} \in \mathrm{R}^q$. Verify the Mahalanobis distance $D_q(\mathbf{X}) = \|\mathbf{Z}\|$, where $\|\cdot\|$ is the Euclidean norm in R^q.

3.29. Let $\boldsymbol{\Sigma}$ be the covariance matrix of $\mathbf{X} \in \mathrm{R}^q$. For $\mathbf{a} \in \mathrm{R}^q$, suppose $\boldsymbol{\Sigma}\mathbf{a} = \lambda_1\mathbf{a}$ with λ_1 being the largest eigenvalue of $\boldsymbol{\Sigma}$. Find a vector $\mathbf{b} \in \mathrm{R}^q$ which maximizes $var(\mathbf{b}^T\mathbf{X})$ subject to $\mathbf{b}^T\mathbf{b} = 1$ and $cov(\mathbf{b}^T\mathbf{X}, \mathbf{a}^T\mathbf{X}) = 0$. *Clearly justify your answer.*

3.30. For a $p \times q$ matrix $\mathbf{M}$, show that $\mathbf{A} = \mathbf{M}\mathbf{M}^{\mathrm{T}}$ and $\mathbf{B} = \mathbf{M}^{\mathrm{T}}\mathbf{M}$ possess the same non-zero eigenvalues. Also, show the eigenspace $\mathcal{E}_B(\lambda) = \{\mathbf{y} \in \mathrm{R}^q \mid \mathbf{B}\mathbf{y} = \lambda\mathbf{y}\} = \{\mathbf{y} = \mathbf{M}^{\mathrm{T}}\mathbf{x} \mid \mathbf{x} \in \mathcal{E}_A(\lambda)\}$, where $\mathcal{E}_A(\lambda) = \{\mathbf{x} \in \mathrm{R}^p \mid \mathbf{A}\mathbf{x} = \lambda\mathbf{a}\}$.